Last time, we started studying statistical inference for linear regression models. After fitting the least squares regression line to estimate the true population relationship between X and Y, we can construct confidence intervals to give a plausible range of values for the slope and for the intercept of the true population model

$y = \beta_0 + \beta_1 x + \varepsilon$ population relationship (unknown)

* estimate $\hat{\beta}_0, \hat{\beta}_1$ from data

* construct confidence intervals for β_0 and β_1

(estimator) $\pm$ (percentile of t_{n-2} distribution)(standard error of estimator)

* hypothesis tests about b_0 and b_1

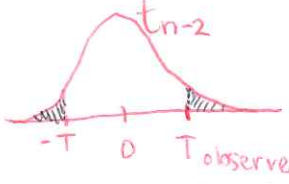
* $H_0: \beta_1 = 0$ $H_a: \beta_1 \neq 0$

slope = 0 means no linear relationship between X and Y.

We can also test hypotheses about the true value of the slope or intercept. One test that arises frequently is the test for whether the true slope is zero. If the slope were truly zero, that would mean there is no linear relationship between x and y. The response variable y tends to take the same values regardless of the value of the predictor x, and knowing x would not help predict y.

$$T = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)}$$

standard deviations away from expected value



we perform the test by calculating the test statistic T, then finding the probability of a value at least that far away from zero, under a t distribution with n-2 degrees of freedom

In addition to performing inference about the slope and intercept of the linear model, we can also perform inference about predictions made by the model, as we will explore today.

next: inference about model predictions

To put this in context, we'll consider an example where we want to predict the amount of water that flows off the surface of a highway when it rains.

ex X = rainfall volume (m^3)

Y = runoff volume (m^3)

Researchers measured the amount of rainfall and the amount of water runoff at a particular highway location on 15 different days.

We assume there is some true underlying linear association between rainfall and runoff at this location.

true assoc. $Y = \beta_0 + \beta_1 X + \varepsilon$ $\quad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

For any particular amount of rainfall, there are two quantities we might want to estimate: the AVERAGE runoff volume on days with that amount of rainfall, and the runoff on one specific future day with that amt. of rainfall.

for particular x^{new}, may want:

A) avg Y value when $X = x^{new}$

$$E(Y|X = x^{new}) = \beta_0 + \beta_1 x^{new}$$

B) specific observation y_i for $x_i = x^{new}$

$$y_i = \beta_0 + \beta_1 x^{new} + \varepsilon_i$$

Suppose we fit a least squares regression line to the observed data, and obtain the following:

estimate: $\hat{y} = -1 + 0.83X$

$\hat{\beta}_0$ $\hat{\beta}_1$

For a day with 100 m^3 of rainfall, if we plug in $X = 100$, we get $\hat{y} = 82\ m^3$.

for $X = 100$, $\hat{y} = -1 + 0.83(100) = 82$

whether we're interested in the average runoff across days with 100 m^3 of rainfall, or the runoff on one specific day when rainfall = 100, we would estimate 82 m^3 of runoff.

A) AVG Y on days w/ X = 100 : predict 82

B) Y on specific day w/ X = 100 : predict 82

} same value

The regression line describes the average Y values at a particular value of X. On any particular day w/ rainfall of 100 m^3 the runoff could be above or below the average, but the average value is still the best guess we can make, since deviations from that average are random & unpredictable.

However, even though our predictions for A & B are the same value, the precision of those two estimates is different. We know neither prediction is exactly correct, since they're estimated from sample data.

precision of the guess (ie how close we are) is different for A vs B

How close we expect those estimates to be to the true values is not the same for A and B.

A: avg is $\mathbb{E}(Y \mid X = 100) = \beta_0 + 100\beta_1$

The true average runoff on days with 100 m^3 of rainfall is given by the Y value on the population regression line for $X = 100$: $\beta_0 + 100\beta_1$.

how close is $\hat{\beta}_0 + \hat{\beta}_1 \times 100$ to $\beta_0 + 100\beta_1$?

Estimating that quantity with the value on our estimated regression line, $\hat{\beta}_0 + 100\hat{\beta}_1 = 82$, we'll be slightly off, because the estimated slope and intercept won't perfectly match the population values (β_0 and β_1).

↳ know precision of $\hat{\beta}_0$ & $\hat{\beta}_1$ estimators

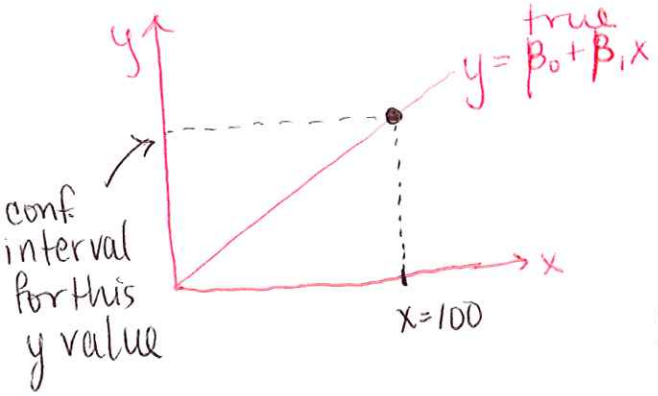
Because we know how far off our estimates of the slope and intercept tend to be from the true values, we can also determine how far off $\hat{\beta}_0 + \hat{\beta}_1 \times 100$ tends to be from $\beta_0 + 100\beta_1$.

↳ use that to determine precision of $\hat{\beta}_0 + 100\hat{\beta}_1$

Using the observed data, that allows us to construct a confidence interval for the quantity $\beta_0 + 100\beta_1$ (or, generically for some value of x, $\beta_0 + \beta_1 x$).

can construct confidence interval for true $\beta_0 + 100\beta_1$

ie true average runoff across all days with rainfall = 100



We are uncertain about this quantity because we don't know the exact slope and intercept of the population relationship.

Suppose we find the 95% confidence interval (instructions later), and it is [78, 86]. We would say:

"95% confident the true average runoff is between 78 and 86 on days with rainfall of 100"

If we're interested in the amount of runoff on one individual day with 100 m^3 of rainfall, the amount could be above or below the average.

(B) runoff on one specific day with 100m^3 of rainfall

↳ on any given day, could be above or below avg

How far above or below do we think it's likely to be, based on the observed data?

population model

$y = \beta_0 + \beta_1 x + \varepsilon$, $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

$y | x=100 \sim N(\beta_0 + 100\beta_1, \sigma^2)$

In the population model, the runoff on a day with ~~runoff~~ RAINFALL =100 is a normally distributed random variable. It is centered at the point ~~on~~ on the population regression line, and has variance σ^2.

on days with $x=100$, y is within $\pm 2\sigma$ of $\beta_0 + 100\beta_1$ with 95% probability

Because the y values are normally distributed, we know the individual values fall within 2 stdev of their average, 95% of the time. If we wanted to give a confidence interval, we only have to account for how spread out the y values are around their average for $x=100$.

how spread out the y values are around their average for $x=100$

BUT we don't know the true average

In practice, we don't know the true average y value when $x=100$. So if we want to give a range of plausible runoff values on some future day with rainfall =100, there is uncertainty because individual days are spread out rand only around the average, and also because we aren't exactly sure what the true average is.

plausible range of runoff values on some future day w/ x=100 :

* avg probably between 78 and 86
* runoff on indiv day w/ x=100 can be below or above avg

↳ maybe 72 to 92

If we made a confidence interval for the runoff on an individual day with rainfall=100, it would be a wider interval than the CI for the AVERAGE runoff on days with rainfall=100, because of the extra variability of individual-day values around the average.

interval for avg ŷ when $x = x^*$: "confidence interval"

interval for one specific y observation when $x = x^*$: "prediction interval"

To distinguish these two types of intervals, the first one is usually referred to as a confidence interval, and the second type as a "prediction interval."

For the same value of x, the prediction interval is wider than the confidence interval.

prediction interval is wider (for any specific x value) (& the same confidence level)

Confidence intervals (for $E(Y|X=x^*)$) and prediction intervals share an interesting property: the width of the interval depends on the value of x where we're making the prediction

interval width (for Ⓐ and Ⓑ) depends on value of x^*

We're most confident about our ability to make predictions near the middle of the observed data, ie $X=\bar{x}$, the sample mean.

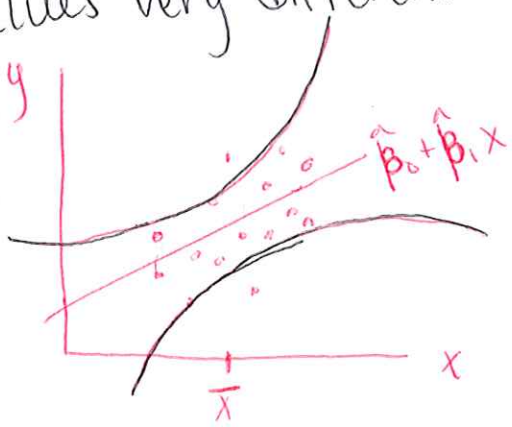
most confident about predictions near middle of data

↳ smallest intervals @ $x^*=\bar{x}$

When we try to make predictions further away from the center of the observed x values, we're less confident, and the intervals are wider.

wider intervals for x^* far away from $\bar{x}$

This connects to our earlier discussion about extrapolation - we aren't confident the model would make good predictions for x values very different from the range of observed values.



(computational examples in notebook)

We've studied how to fit linear models, how to interpret them, and how to use them for making predictions. The next topic to consider is how we assess whether the model is good.

next: assessing model quality

One starting question is whether the fitted model is successful at predicting y values for a given x value.

is fitted model successful at predicting y from x?

Each observation (x_i, y_i) has an associated random error term, so it isn't possible to perfectly predict y from x. But how close do the predictions tend to be?

can't perfectly predict y due to random error

$\sqrt{\hat{\sigma}^2}$ describes how close the predictions tend to be

The standard error of the residuals is one measure we can consider. It describes how far away the actual response y_i is from the fitted $\hat{y}_i$, on average.

If the y values tend to be more spread out around the true population regression line, then no matter how well we estimate the regression line, the $\hat{y}_i$ values we predict can be quite far off.

If we want to describe how useful the linear model is for predicting y from x, another measure we use is called R^2.

another measure: R^2

For linear regression with only one predictor variable x, R^2 is literally the square of the sample correlation, r, between x and y. (the observed x & y data values, that is)

for simple linear regression: r_{xy} = sample correlation btwn x, y

$$R^2 = (r_{xy})^2$$

$$r_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

The correlation can take values from negative one to positive one, and R^2 ranges from 0 to 1.

$$-1 \leq r_{xy} \leq 1 \qquad 0 \leq R^2 \leq 1$$

If $R^2 = 0$, there is no correlation between the observed x and y values. A linear regression model would not help predict y, because the values of x and y are not correlated.

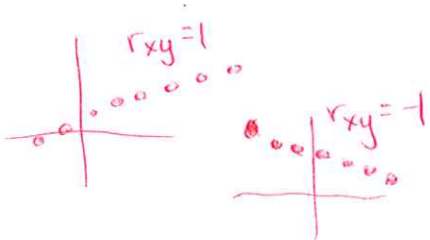
$R^2 = 0$ no correlation

linear model not useful for predicting y

If x and y are perfectly positively correlated OR perfectly negatively correlated, $R^2 = 1$. In that case, the linear model allows us to exactly predict the value of y from the value of x. $r_{xy} = 1$ or $r_{xy} = -1 \Rightarrow R^2 = 1$

perfect correlation

linear model exactly predicts y



In between those two extremes, larger values of R^2 indicate that the linear model is relatively more useful for predicting y from x.

generally: larger R^2 → linear model more successful at predicting y

Why bother making the distinction and talking about R^2 instead of the correlation? R^2 has a special interpretation that will also be useful later on when we study linear models with multiple predictor variables instead of just one.

why talk about R^2? has special interpretation that applies beyond simple linear regression

And that interpretation is that R^2 also equals...

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

numerator: variability of y values around fitted line

denominator: variability in observed y values

The numerator is the sum of squared residuals, and the denominator quantifies how spread out the observed y values were, overall (inherent variability in y, overall before we fit a regression model).

The sum of squared residuals tells us how spread out the observed y values are around the fitted regression line. In other words, that is variability in y that remains after fitting the model.

$R^2 = 1 - \frac{\text{variation in } y \text{ NOT explained by model}}{\text{total variation in } y}$

So conceptually, R^2 equals 1 minus the FRACTION of the total variability in y that is NOT explained by the model.

= 1 - fraction of y variability not explained by model

= Fraction of y variability that IS explained by the model

R^2 tells us how much (what perantage) of the overall variability in the observed y values is explained by x in the fitted linear model.

all data points on line → all residuals $y_i - \hat{y}_i = 0$

$$R^2 = 1 - \frac{0}{\Sigma(y_i - \bar{y})^2} = 1$$

If all the observed data points fell perfectly on a line, the residuals would all equal zero, and R^2 would equal 1. The fitted linear model would fully explain all of the variation in the y values.

in general: Larger R^2 → linear model explains more of the variation in y

& is more useful for predicting y

A small R^2 value tells us that the linear model is not very successful at predicting y.

small R^2 → linear model not very good at predicting y

Why might that happen?

One possibility is that x and y have a linear association, but it isn't very strong.

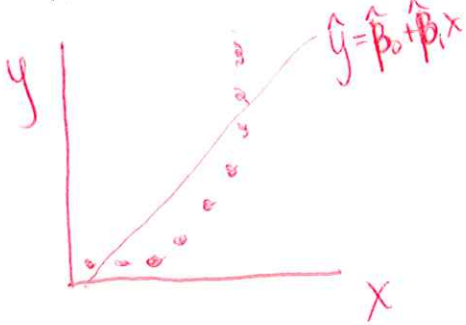
① linear relationship, but weak association

Another possibility is that the linear regression model we fit was not a good choice.

② wrong model

Something important to be aware of is that the correlation, r_{xy}, and R^2, measure the strength of linear relationships.

If x and y are associated, but the relationship isn't linear, then the correlation value may be small, and a linear model may do a bad job predicting the value of y from the value of x.



NONlinear associations also exist

